LABORATORIO DOCKER

Construir una imagen desde Dockerfile

```
C:\Users\ThinkPad\docker-lab>docker build -t miapp:v1 .
[+] Building 5.7s (8/8) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                  0.2s
 => => transferring dockerfile: 149B                                                                  0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                        2.3s
 => [auth] library/alpine:pull token for registry-1.docker.io                                         0.0s
 => [internal] load .dockerignore                                                                     0.1s
 => => transferring context: 2B                                                                       0.0s
 => [1/3] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03  1.2s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03  0.1s
 => => sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd 3.63MB / 3.63MB        0.8s
 => => extracting sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd             0.3s
 => [2/3] WORKDIR /app                                                                                0.1s
 => [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt                                             0.6s
 => exporting to image                                                                                0.7s
 => => exporting layers                                                                               0.3s
 => => exporting manifest sha256:9d3f31a5ecaa3e6146a8d28bf8c30b4b8a1330d0d562972fc625e89b129fc757     0.0s
 => => exporting config sha256:99d140cc254104c4624d634322f15693215405a5c0ddaf727aa3d2e4ad40a02c       0.0s
 => => exporting attestation manifest sha256:bf95880d845d2fa8465528ec772d708104fcfd6cfef2ff240f45f09e87b19fed  0.1s
 => => exporting manifest list sha256:f6cd3e3a0869834808b98046aa5c1c48fe07c6db7de956d42baa7fa63f9a9bc6  0.0s
 => => naming to docker.io/library/miapp:v1                                                           0.0s
 => => unpacking to docker.io/library/miapp:v1                                                        0.1s
```

Comprobando que este funcionando

```
C:\Users\ThinkPad\docker-lab>docker run --rm miapp:v1
Hola desde Docker
```

Etiquetar una imagen

		Name	Tag	Image ID	Created	Size	Actions
☐	○	miapp	latest	f6cd3e3a0869	6 minutes ago	12.11 MB	
☐	○	miapp	v1	f6cd3e3a0869	6 minutes ago	12.11 MB	

Inspeccionar una imagen

```
C:\Users\ThinkPad\docker-lab>docker image inspect miapp:v1 --format '{{.Id}}'
'sha256:f6cd3e3a0869834808b98046aa5c1c48fe07c6db7de956d42baa7fa63f9a9bc6'
```

Docker Container

```
C:\Users\ThinkPad\docker-lab>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS
          NAMES
d6f233cb4ff7   nginx:latest   "/docker-entrypoint.…"   25 seconds ago   Up 23 seconds   0.0.0.0:8080->80/tcp, [::]:8080-
>80/tcp   web2
5647c568555a   nginx:latest   "/docker-entrypoint.…"   3 minutes ago    Up 3 minutes    80/tcp
          web1
```

Ejecutar contenedor con nombre y puertos

localhost:8080

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

Listar todos los contenedores

```
C:\Users\ThinkPad\docker-lab>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS
        NAMES
d6f233cb4ff7   nginx:latest   "/docker-entrypoint.…"   2 minutes ago   Up 2 minutes   0.0.0.0:8080->80/tcp, [::]:8080->8
0/tcp   web2
5647c568555a   nginx:latest   "/docker-entrypoint.…"   5 minutes ago   Up 5 minutes   80/tcp
        web1
```

Ver logs

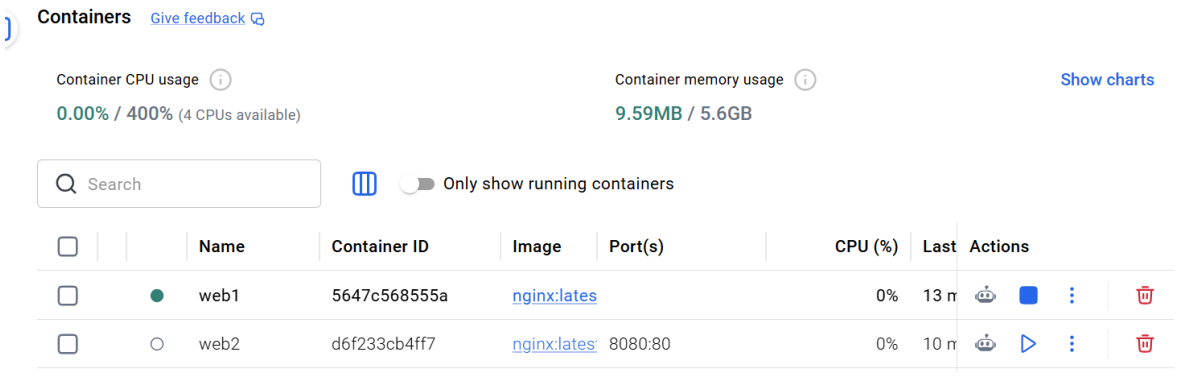
```
C:\Users\ThinkPad\docker-lab>docker logs web2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/11 20:07:44 [notice] 1#1: using the "epoll" event method
2026/04/11 20:07:44 [notice] 1#1: nginx/1.29.8
2026/04/11 20:07:44 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/04/11 20:07:44 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/04/11 20:07:44 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/11 20:07:44 [notice] 1#1: start worker processes
2026/04/11 20:07:44 [notice] 1#1: start worker process 29
2026/04/11 20:07:44 [notice] 1#1: start worker process 30
2026/04/11 20:07:44 [notice] 1#1: start worker process 31
2026/04/11 20:07:44 [notice] 1#1: start worker process 32
172.17.0.1 - - [11/Apr/2026:20:08:22 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) Appl
eWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36" "-"
2026/04/11 20:08:22 [error] 30#30: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory),
client: 172.17.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referrer: "http://l
ocalhost:8080/"
172.17.0.1 - - [11/Apr/2026:20:08:22 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (W
indows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36" "-"
```

Inspeccionar un contenedor

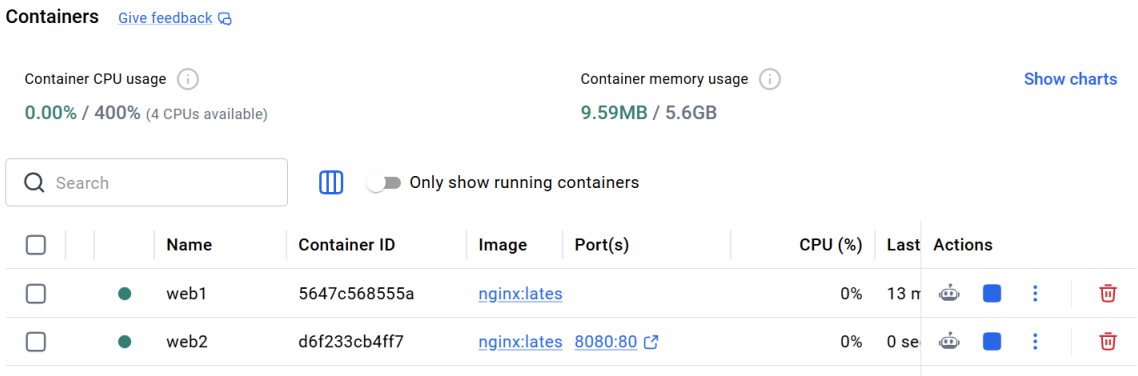
```
C:\Users\ThinkPad\docker-lab>docker inspect web2 --format '{{.State.Status}}'
'running'
```

Detener y arrancar contenedor

```
C:\Users\ThinkPad\docker-lab>docker stop web2
web2
```




```
C:\Users\ThinkPad\docker-lab>docker start web2
web2
```



Entrar en modo interactivo

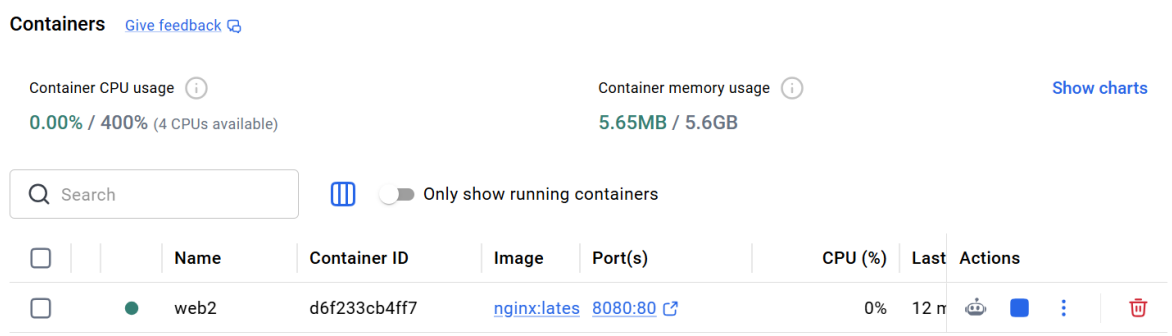
```
C:\Users\ThinkPad\docker-lab>docker run -it --name alpine1 alpine:3.20 sh
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.9
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ #
```

Pausar y reanudar

```
C:\Users\ThinkPad\docker-lab>docker pause web2
web2

C:\Users\ThinkPad\docker-lab>docker inspect web2 --format '{{.State.Paused}}'
'true'

C:\Users\ThinkPad\docker-lab>docker unpause web2
web2

C:\Users\ThinkPad\docker-lab>docker inspect web2 --format '{{.State.Paused}}'
'false'
```

Eliminar contenedores

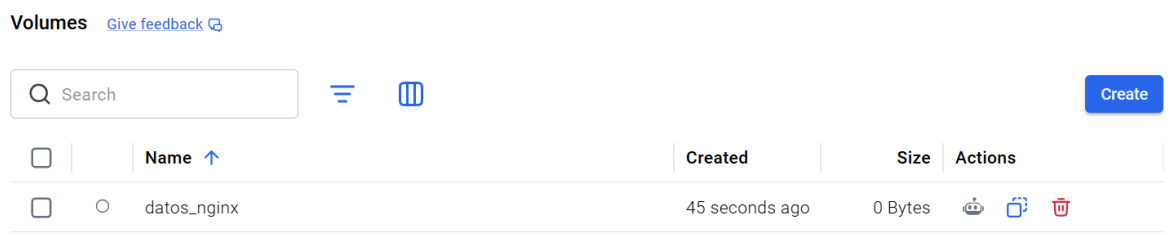
```
C:\Users\ThinkPad\docker-lab>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS
           NAMES
d6f233cb4ff7   nginx:latest   "/docker-entrypoint.…"   21 minutes ago   Up 11 minutes   0.0.0.0:8080->80/tcp, [::]:8080-
>80/tcp   web2
```



Volúmenes

Crear volumen

```
C:\Users\ThinkPad\docker-lab>docker volume create datos_nginx
datos_nginx

C:\Users\ThinkPad\docker-lab>docker volume ls
DRIVER    VOLUME NAME
local     datos_nginx
```



Inspeccionar volumen

```
C:\Users\ThinkPad\docker-lab>docker volume inspect datos_nginx
[
    {
        "CreatedAt": "2026-04-11T20:35:07Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/datos_nginx/_data",
        "Name": "datos_nginx",
        "Options": null,
        "Scope": "local"
    }
]
```

Usar Volumen

```
C:\Users\ThinkPad\docker-lab>echo "<h1>Hola volumen</h1>" > index.html

C:\Users\ThinkPad\docker-lab>dir
 El volumen de la unidad C no tiene etiqueta.
 El número de serie del volumen es: B41E-BEAA

 Directorio de C:\Users\ThinkPad\docker-lab

11/04/2026  03:38p. m.    <DIR>          .
11/04/2026  02:43p. m.    <DIR>          ..
11/04/2026  02:53p. m.               112 Dockerfile
11/04/2026  03:38p. m.                26 index.html
11/04/2026  03:03p. m.         3.645.440 miapp.tar
               3 archivos      3.645.578 bytes
               2 dirs  758.136.770.560 bytes libres

C:\Users\ThinkPad\docker-lab>docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to webvol:/usr/share/nginx/html/index.html
```

"

Hola volumen

"

Eliminar Volumen

```
C:\Users\ThinkPad\docker-lab>docker rm -f webvol
webvol

C:\Users\ThinkPad\docker-lab>docker volume rm datos_nginx
datos_nginx

C:\Users\ThinkPad\docker-lab>docker volume ls
DRIVER    VOLUME NAME
```

Volumes Give feedback

Containers can use volumes to store data

All data in a container is lost once it is removed. Containers use volumes to persist data.

Create a volume

Redes

Listar redes

```
C:\Users\ThinkPad\docker-lab>docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
4ce9e11a17f6   bridge    bridge    local
02efda93dcbc   host      host      local
1a41919580b1   none      null      local
```

Crear Red (mi_red)

```
C:\Users\ThinkPad\docker-lab>docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
4ce9e11a17f6   bridge    bridge    local
02efda93dcbc   host      host      local
5b087f2084a3   mi_red    bridge    local
1a41919580b1   none      null      local
```

Conectar contenedores a una red

```
C:\Users\ThinkPad\docker-lab>docker run -d --name caja1 --network mi_red alpine:3.20 sleep 300
a16755b6a1e51d5afdb4292c06f6a041aa49b349f3a29017b5ca83f268290321

C:\Users\ThinkPad\docker-lab>docker run -d --name caja2 --network mi_red alpine:3.20 sleep 300
62a77cf07642dbc11d3a90c95d916fa58306cf18b2ff5018acae7e68aa3b8d25

C:\Users\ThinkPad\docker-lab>docker exec caja1 ping -c 2 caja2
PING caja2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.640 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.087 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
```

Desconectar un contenedor de la red

```
C:\Users\ThinkPad\docker-lab>docker network disconnect mi_red caja2

C:\Users\ThinkPad\docker-lab>docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'
```

Docker Compose

Levantar servicios

```
C:\Users\ThinkPad\docker-lab>docker compose up -d
[+] up 2/2
 ✔Network docker-lab_default Created                                                                    0.2s
 ✔Container docker-lab-web-1 Started                                                                    0.8s

C:\Users\ThinkPad\docker-lab>docker compose ps
NAME                IMAGE          COMMAND                  SERVICE   CREATED          STATUS          PORTS
docker-lab-web-1    nginx:latest   "/docker-entrypoint.…"   web       33 seconds ago   Up 32 seconds   0.0.0.0:8082->80/t
cp, [::]:8082->80/tcp
```

Ver Logs

```
C:\Users\ThinkPad\docker-lab>docker compose logs
web-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1  | 2026/04/11 20:51:20 [notice] 1#1: using the "epoll" event method
web-1  | 2026/04/11 20:51:20 [notice] 1#1: nginx/1.29.8
web-1  | 2026/04/11 20:51:20 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1  | 2026/04/11 20:51:20 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
web-1  | 2026/04/11 20:51:20 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1  | 2026/04/11 20:51:20 [notice] 1#1: start worker processes
web-1  | 2026/04/11 20:51:20 [notice] 1#1: start worker process 29
web-1  | 2026/04/11 20:51:20 [notice] 1#1: start worker process 30
web-1  | 2026/04/11 20:51:20 [notice] 1#1: start worker process 31
web-1  | 2026/04/11 20:51:20 [notice] 1#1: start worker process 32
web-1  | 172.19.0.1 - - [11/Apr/2026:20:53:06 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safar
i/537.36 Edg/146.0.0.0" "-"
web-1  | 2026/04/11 20:53:06 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.19.0.1, server: localhost, request: "GET /fav
icon.ico HTTP/1.1", host: "localhost:8082", referrer: "http://localhost:8082/"
web-1  | 172.19.0.1 - - [11/Apr/2026:20:53:06 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, li
ke Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0" "-"
web-1  | 172.19.0.1 - - [11/Apr/2026:20:53:18 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safar
i/537.36" "-"
web-1  | 2026/04/11 20:53:18 [error] 31#31: *3 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.19.0.1, server: localhost, request: "GET /fav
icon.ico HTTP/1.1", host: "localhost:8082", referrer: "http://localhost:8082/"
web-1  | 172.19.0.1 - - [11/Apr/2026:20:53:18 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, li
ke Gecko) Chrome/147.0.0.0 Safari/537.36" "-"
```

Bajar Servicios

```
C:\Users\ThinkPad\docker-lab>docker compose down
[+] down 2/2
 ✔Container docker-lab-web-1 Removed
       0.8s
 ✔Network docker-lab_default Removed
       0.3s

C:\Users\ThinkPad\docker-lab>docker compose ps
NAME      IMAGE     COMMAND   SERVICE   CREATED   STATUS    PORTS
```

Escalar servicio

```
C:\Users\ThinkPad\docker-lab>docker compose up -d --scale web=3
[+] up 3/4
 ✔Network docker-lab_default Created                                                                    0.1s
 - Container docker-lab-web-3 Starting                                                                  1.7s
 ✔Container docker-lab-web-1 Started                                                                    1.1s
 ✔Container docker-lab-web-2 Created                                                                    0.3s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint do
cker-lab-web-3 (92e40d701ea8a12ef13ac34514f1bb01bfe487b1bf6a1ef954be6c50e4347da6): Bind for 0.0.0.0:8082 failed: port is already
allocated
```

Punto 6

Construir contenedor

```
C:\Users\ThinkPad\dockerfile-test>docker build -t dockerfile-jose --build-arg VERSION=2.0 .
[+] Building 2.5s (10/10) FINISHED                                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                    0.1s
 => => transferring dockerfile: 389B                                                                    0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                          0.1s
 => [internal] load .dockerignore                                                                       0.1s
 => => transferring context: 2B                                                                         0.0s
 => [internal] load build context                                                                       0.1s
 => => transferring context: 99B                                                                        0.0s
 => [1/5] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805   0.1s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805   0.1s
 => CACHED [2/5] WORKDIR /app                                                                           0.0s
 => [3/5] COPY archivo.txt /app/archivo.txt                                                             0.1s
 => [4/5] RUN echo "Version del Proyecto: 2.0" > /app/version.txt                                       0.7s
 => [5/5] ADD archivo.txt /app/archivo2.txt                                                             0.1s
 => exporting to image                                                                                  0.8s
 => => exporting layers                                                                                 0.4s
 => => exporting manifest sha256:b0605c756a6b8c44eb44e0875addc740c527bf879fa1e7797e2e569b88cd8168       0.0s
 => => exporting config sha256:efb869c188ce512dfdac1d96295ae021b61cd66dfd1b248217cf436b833be256         0.0s
 => => exporting attestation manifest sha256:c47b80ec4e33087cd7bac69d3063364e7a1abeddfa19c7f6235eae3ecbfb3a57   0.1s
 => => exporting manifest list sha256:642ffc7741fe2b8d5bbdbe6ed7ea1acad2f27c139926fd3a5740a195ee945107   0.0s
 => => naming to docker.io/library/dockerfile-jose:latest                                               0.0s
 => => unpacking to docker.io/library/dockerfile-jose:latest                                            0.2s
```

Comprobar

```
C:\Users\ThinkPad\dockerfile-test>docker run --rm dockerfile-jose
Entorno: desarrollo
echo "Este es el proyecto de José Manuel Cruz" > archivo.txtVersion del Proyecto: 2.0
```